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Notation

$\mathbb{N} = \{1, 2, 3, \dots\}$	set of natural numbers (positive integers)
$\mathbb{Z}_n = \{0, 1, \dots, n-1\}$	ring of residue classes modulo n
$\mathbb{Z}, \mathbb{Q}, \mathbb{R}, \mathbb{C}$	sets of integer, rational, real and complex numbers
$]a, b[$ and $[a, b]$	open and closed interval from a to b
$\llbracket a, b \rrbracket$	set of integers from a to b
$\mathbb{Z}[i]$	ring of Gaussian integers $\mathbb{Z}[i] = \{a + bi \mid a, b \in \mathbb{Z}\}$
$\mathbb{Z}^{n \times n}$	set of integer square matrices of dimension n
$\mathbb{Z}[t]$ and $\mathbb{Q}[t]$	rings of polynomials with integer and rational coefficients
$ x $	absolute value of x
$\#M$ or $ M $	cardinality of a set M
$M \times N$	Cartesian product of two sets or graphs M and N
$M \cup N, M \cap N, M \setminus N$	union, intersection and difference of two sets M and N

1. Prefix Sequences

Let $(a_k) = a_1, a_2, \dots, a_k, \dots$ be a sequence of positive integers, and let $n \in \mathbb{N}$ be a positive integer. The *n-prefix* of (a_k) is the sequence (b_k) constructed as follows: the number b_k is the smallest positive integer such that $\overline{b_k a_k}$ is divisible by n , where $\overline{x}$ denotes the decimal expansion of x . For example, if $(a_k) = 1, 2, 3, 4, 5, 6, \dots$ is the sequence of natural numbers, then its 3-prefix is the periodic sequence $2, 1, 3, 2, 1, 3, \dots$

Try to answer the questions 1–3 below when (a_k) is:

- a) the sequence of natural numbers $1, 2, 3, 4, 5, \dots$;
- b) an arithmetic sequence: $a_{k+1} = a_k + d$ for some $d \in \mathbb{N}$ and all $k \in \mathbb{N}$;
- c) a geometric sequence: $a_{k+1} = a_k \cdot q$ for some $q \in \mathbb{N}$ and all $k \in \mathbb{N}$;
- d) the Fibonacci sequence $1, 1, 2, 3, 5, \dots$;
- e) a recursive sequence: $a_{k+2} = \alpha a_{k+1} + \beta a_k$ for some $\alpha, \beta \in \mathbb{Z}$ and all $k \in \mathbb{N}$;
- f) the sequence of primes $2, 3, 5, 7, 11, \dots$;
- g) another sequence of your choice;
- h) an arbitrary sequence.

1. Determine whether the n -prefix of (a_k) contains all integers from 1 to n .
2. Determine whether the n -prefix of (a_k) is periodic.
3. Find a formula for the n -prefix of (a_k) .
4. Find necessary and/or sufficient conditions for two sequences to have the same n -prefix, if these two sequences are:
 - a) both arithmetic (with differences d_1 and d_2);
 - b) both geometric (with ratios q_1 and q_2);
 - c) an arithmetic (with difference d) and a geometric (with ratio q);
 - d) both recursive (with coefficients α_1, β_1 and α_2, β_2);
 - e) arbitrary.
5. Introduce the notion of the n -postfix of (a_k) and study similar questions.
6. Suggest and investigate other directions of research.

2. Free Triangulations

A set $S = \{P_1, \dots, P_{3n}\}$ of $3n$ points in the real plane is given, and no three of these points are on the same line. A *free triangulation* of S is its partition into n triples of points such that the triangles with vertices at the triples are pairwise disjoint (the triangles don't have common inner or boundary points). See examples in Figure 1.

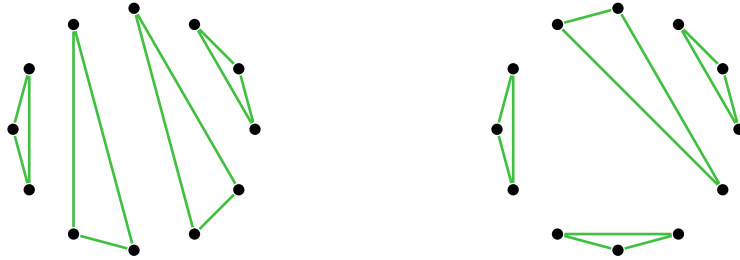


FIGURE 1. Two free triangulations of the set of vertices of a regular dodecagon.

Denote by $f_3(S)$ the number of distinct free triangulates of S . (Two different partitions of S give distinct triangulations.)

1. Prove that S has at least one free triangulation.
2. Show that the inequality $f_3(S) \geq 3$ is true for any set S when $n \geq 2$.
3. Try to find a better lower bound for $f_3(S)$ in terms of n .
4. Find a formula for $f_3(S)$ when S is the set of vertices of a regular $3n$ -gon.
5. Find a formula for $f_3(S)$ for other particular families of points S .
6. Find upper bounds for $f_3(S)$ in terms of n .

7. Given n , describe the sets S of $3n$ points for which $f_3(S)$ attains its maximum value.
8. In general, let $k \geq 3$ and n be positive integers, and let S be a set of kn points in the real plane with no three on the same line. Consider the number $f_k(S)$ of partitions of S into k -tuples such that one may draw k -gons with vertices at the k -tuples, that don't intersect each other. Find lower and upper bounds for $f_k(S)$. Try to find a formula for particular sets S .
9. Let S be a set of n points in the real plane with no three on the same line. Let $n = a_1 + \dots + a_k$ be a decomposition of n into a sum of positive integers more than 2. Does there exist a partition of S into an a_1 -tuple, $\dots$, an a_k -tuple so that the k polygons with vertices at these tuples might be constructed to be **convex** and pairwise disjoint? Give an algorithm that checks this and finds such a partition if it exists.
10. Suggest and investigate additional directions of research.

3. Towers of Bricks

Let n and $k \leq n$ be positive integers. A *tower of n bricks with k floors* is a sequence of k positive integers $(u_1, u_2, \dots, u_k)$ such that $u_1 + u_2 + \dots + u_k = n$ and

$$u_{i+1} \geq u_1 + \dots + u_i \quad \text{for all } 1 \leq i \leq k-1.$$

Denote by $a(n, k)$ the number of towers of n bricks with k floors, and let $a(n) = \sum_{k=1}^n a(n, k)$ be the number of all towers of n bricks.

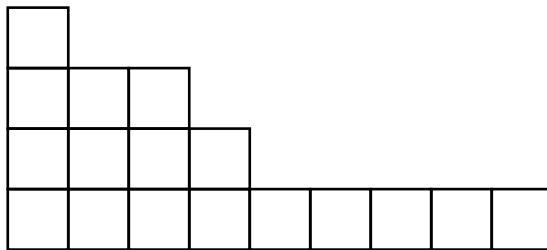


FIGURE 2. A tower of 17 bricks with 4 floors ($u_1 = 1, u_2 = 3, u_3 = 4, u_4 = 9$).

1. Determine the maximum number of floors that a tower of n bricks can have.
2. Give a formula in terms of n or find bounds for
 - a) $a(n, 2)$;
 - b) $a(n, 3)$;
 - c) $a(n, k)$ with $k > 3$.
3. Find lower/upper bounds and an asymptotic for $a(n)$.
4. Find necessary and/or sufficient conditions on n so that
 - a) $a(n)$ is even;
 - b) $a(n)$ is divisible by 2^b for a positive integer $b > 1$.
5. Let m and n be positive integers. Find necessary and/or sufficient conditions on n so that
 - a) $a(m) \equiv a(n)$ modulo 4;
 - b) $a(m) \equiv a(n)$ modulo 2^b for a positive integer $b > 2$.
6. In general, given a prime p and a positive integer b , describe all numbers $n \in \mathbb{N}$ such that $a(n)$ is divisible by p^b .
7. Suggest and study additional directions of research.

4. Maximizing Distances

Consider the Euclidean plane $\mathbb{R}^2$. The Euclidean distance is the function $d_2 : \mathbb{R}^2 \times \mathbb{R}^2 \rightarrow \mathbb{R}$ such that

$$d_2((x_1, y_1), (x_2, y_2)) = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}.$$

Given n -points with coordinates $(x_1, y_1), \dots, (x_n, y_n) \in \mathbb{R}^2$, let us denote by S_2 the sum of all pairwise distances between them:

$$S_2((x_1, y_1), \dots, (x_n, y_n)) = \sum_{1 \leq i < j \leq n} d_2((x_i, y_i), (x_j, y_j)).$$

See an example of such a sum for four points in Figure 3.

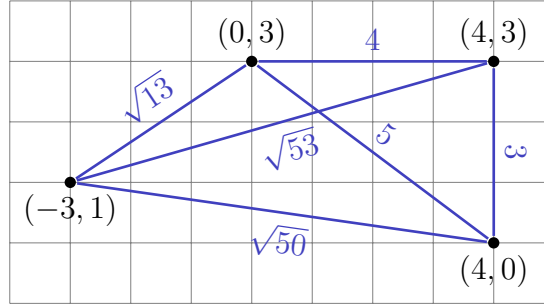


FIGURE 3. $S_2((-3, 1), (0, 3), (4, 0), (4, 3)) = 12 + \sqrt{13} + \sqrt{50} + \sqrt{53}$.

In this problem, Ω will stand for a convex compact subset of $\mathbb{R}^2$, that is:

- for any two points $(x_1, y_1), (x_2, y_2) \in \Omega$ and any $t \in [0, 1]$, one has

$$(tx_1 + (1-t)x_2, ty_1 + (1-t)y_2) \in \Omega;$$

- for any sequence of points $(x_1, y_1), \dots, (x_n, y_n), \dots$ from Ω , if both $\lim_{n \rightarrow \infty} x_n$ and $\lim_{n \rightarrow \infty} y_n$ exist, then the point $(\lim_{n \rightarrow \infty} x_n, \lim_{n \rightarrow \infty} y_n)$ is also in Ω ;
- Ω is contained in a disc of $\mathbb{R}^2$.

We are interested in finding the following expression:

$$M_2(\Omega, n) = \max \left\{ S_2((x_1, y_1), \dots, (x_n, y_n)) \mid (x_1, y_1), \dots, (x_n, y_n) \in \Omega \right\}$$

which is the maximum value taken by S_2 on a set of n points from Ω .

For a positive integer d , denote by Ω_d the full regular d -gon whose vertices are on the circle of radius 1. For instance, Ω_1 is a point, Ω_2 is a line segment of length 2 and Ω_3 is a regular triangle with side length $\sqrt{3}$.

1. Check that Ω_d is a convex compact subset.

2. Given $n \in \mathbb{N}$, determine $M_2(\Omega, n)$ when:

- $\Omega = \Omega_2$;
- $\Omega = \Omega_3$;
- $\Omega = \Omega_d$ for any $d > 3$;
- Ω is a disc of radius 1;
- $\Omega = \Omega_{a,b} = \{(x, y) \in \mathbb{R}^2, \frac{x^2}{a^2} + \frac{y^2}{b^2} \leq 1\}$ is an ellipse with width $2a > 0$ and height $2b > 0$.

3. For each of the above subsets, describe the configurations of points $(x_1, y_1), \dots, (x_n, y_n) \in \Omega$ for which the maximum $M_2(\Omega, n)$ is reached.

4. Among the convex compact subsets of $\Omega \subset \mathbb{R}^2$ of area 1, which of them have the largest value of $M_2(\Omega, n)$?

5. Given d , find all positive integers $n \in \mathbb{N}$ for which there exist $n + 1$ points $(x_1, y_1), \dots, (x_{n+1}, y_{n+1}) \in \Omega_d$ satisfying the two following relations at the same time:

$$\begin{cases} S_2((x_1, y_1), \dots, (x_n, y_n)) = M_2(\Omega_d, n), \\ S_2((x_1, y_1), \dots, (x_{n+1}, y_{n+1})) = M_2(\Omega_d, n + 1). \end{cases}$$

6. Given n , describe the convex compact subsets of $\Omega \subset \mathbb{R}^2$ which contain $n + 1$ points $(x_1, y_1), \dots, (x_{n+1}, y_{n+1})$ satisfying the two following relations at the same time:

$$\begin{cases} S_2((x_1, y_1), \dots, (x_n, y_n)) = M_2(\Omega, n), \\ S_2((x_1, y_1), \dots, (x_{n+1}, y_{n+1})) = M_2(\Omega, n + 1). \end{cases}$$

7. In the previous questions, what happens if we replace the function S_2 by the function S_p such that

$$S_p((x_1, y_1), \dots, (x_n, y_n)) = \sum_{1 \leq i < j \leq n} \sqrt[p]{|x_i - y_i|^p + |x_j - y_j|^p}$$

for a parameter $p \in [1, +\infty[$?

8. Investigate a similar problem in $\mathbb{R}^3$.

9. Suggest and investigate additional directions of research.

5. Random Walks on Finite Graphs

Let $G = (V, E)$ be a simple finite graph drawn in the real plane, where V is its set of vertices and E is its set of edges. Two vertices are said to be *neighbours* or *neighbouring* if they are connected by an edge. In the beginning, a beetle is situated at some vertex a of the graph. Every second it either moves to one of the neighbouring vertices or stays where it is, all with equal probabilities. For instance, if G is the 4×4 square grid and a is the lower left corner vertex as in Figure 4, then after 1 second it can still be at the corner with probability $1/3$ or at one of its two neighbours with probabilities $1/3$. Denote by $P_a^n(v)$ the probability that after n seconds the beetle will be at the vertex v .



FIGURE 4. A 4×4 square grid with a beetle at the lower left corner a , and two vertices v and w such that $v \leq_a w$.

For two vertices $v, w \in V$, we will write $v \leq_a w$ if there exists a shortest path from a to w that passes through v (see Figure 4).

1. Let G be the $m \times k$ square grid for some natural numbers m and k that are the lengths of the sides of the grid. Determine all pairs of vertices v and w such that

$$v \leq_a w \quad \text{and} \quad P_a^n(v) \geq P_a^n(w). \quad (1)$$

Consider the following cases:

- a) $m = k = 3$, $n = 6$ and a is the lower left corner vertex of G ;
- b) $m = k = 5$, $n = 10$ and a is the lower left corner vertex of G ;
- c) $m = k = 3$ or 5 , the vertex a is at the lower left corner of G but n is arbitrary;
- d) $m = k = 3$ or 5 but a and n is arbitrary;

- e) m, k, n and a are arbitrary.
2. Find all vertices v and w satisfying (1) for a given starting vertex a when G is
- a finite triangular grid;
 - an infinite triangular grid;
 - a finite hexagonal grid.
 - an infinite hexagonal grid.
 - a regular tree of degree d (the infinite tree with d edges at each vertex).
3. Investigate an analogous problem when G is a 3-dimensional grid $m \times k \times \ell$.
4. A *permutohedron* A_d of order d is the polytope in the d -dimensional space $\mathbb{R}^d$ with $d!$ vertices formed by permuting coordinates of the point $(1, 2, \dots, d)$ and such that two vertices $(a_1, \dots, a_d)$ and $(b_1, \dots, b_d)$ are connected by an edge if and only if one can be obtained from the other by swapping two coordinates, whose values differ by 1. Investigate the analogous problem for the permutohedron A_d .
5. Now, suppose that the beetle cannot stay where it is and always moves to one of the neighbouring vertices. Try to understand how it affects the ordering of probabilities.
6. Suggest and study additional directions of research.

6. Integration of Sequences

Let X be a nonempty set of real numbers, and let W be a family of subsets of X such that $\emptyset \in W$ and $X \in W$. Let $p : W \rightarrow [0, +\infty[$ be an *additive function on W* , that is, for any element $E \in W$ and any partition $\{E_i\}_{i=1}^n$ of E with $n \in \mathbb{N}$ and $E_1, \dots, E_n \in W$, one has

$$p(E) = \sum_{i=1}^n p(E_i).$$

(A *partition* of a set is a collection of its subsets which are pairwise disjoint and whose union is the entire set. For example, $\{\{1, 2\}, \{3\}, \{4, 5, 6\}\}$ is a partition of the set $\{1, 2, 3, 4, 5, 6\}$.)

The triple (X, W, p) will be called *admissible* if, for any bounded real function $f : X \rightarrow \mathbb{R}$, any partitions $\{E_i\}_{i=1}^n \subseteq W$ and $\{D_j\}_{j=1}^k \subseteq W$ of X and any real numbers $m_1, \dots, m_n, M_1, \dots, M_k$, the condition that

$$f(E_i \cap D_j) \subseteq [m_i, M_j] \text{ for all } 1 \leq i \leq n \text{ and } 1 \leq j \leq k \quad (2)$$

implies the inequality

$$\sum_{i=1}^n m_i p(E_i) \leq \sum_{j=1}^k M_j p(D_j).$$

Moreover, if for such a function f , there exists a unique number I such that

$$\sum_{i=1}^n m_i p(E_i) \leq I \leq \sum_{j=1}^k M_j p(D_j)$$

for any E_i, D_j, m_i, M_j as above satisfying (2), then f is said to be *Riemann integrable on (X, W, p)* and I is called its *integral*.

The pair (X, W) is said to be a *Riemann pair* if the triple (X, W, p) is admissible for any additive function p on W .

1. Denote by W^* the set W excluding the trivial elements $\emptyset$ and X . Determine whether (X, W) is a Riemann pair in the following cases:

- $X = \{1, 2, 3, 4, 5, 6\}$ and $W^* = \{\{1, 2\}, \{2, 3\}, \{3, 4\}, \{4, 5\}, \{5, 6\}, \{6, 1\}\}$;
- X is a finite set and W^* is the family of all its subsets of cardinality 2;

- c) X is a finite set and W^* is the family of all its subsets of cardinality $\ell \in \mathbb{N}$;
- d) X is a finite set and W^* is the family of all its nonempty subsets of cardinality at most ℓ .

2. Suppose that $X = \mathbb{N}$ is the set of natural numbers, and let W be the family of all subsets $E \subseteq X$ such that either E or $X \setminus E$ is finite.

- a) Is (X, W) a Riemann pair?
- b) Let $p : W \rightarrow [0, +\infty[$ be the function such that, for any $E \in W$, one has $p(E) = 0$ if E is finite and $p(E) = 1$ if $X \setminus E$ is finite. Find all $d \in \mathbb{N}$ for which the function $f(x) = \sin(x^d)$ and/or $f(x) = \sin(d^x)$ is Riemann integrable on (X, W, p) ?
- c) Is it true that a bounded function $f : X \rightarrow \mathbb{R}$ is Riemann integrable on (X, W, p) if and only if $\lim_{n \rightarrow +\infty} f(n)$ exists?

3. Suppose that $X = \mathbb{N}$ is the set of natural numbers, and let W be the family of all subsets $E \subseteq X$ for which the following limit exists (also called the *density* of E):

$$p(E) = \lim_{n \rightarrow +\infty} \frac{|E \cap [1, n]|}{n}. \quad (3)$$

- a) Show that the function p defined in (3) is additive on W .
- b) Let $A, B \in W$. Is it true that $A \cap B$, $A \cup B$ and/or $A \setminus B$ also belong to W ?
- c) Let $B \in W$. Is it true that, for any number $\alpha \in [0, p(B)]$, there exists an element $A \in W$ such that $A \subseteq B$ and $p(A) = \alpha$?
- d) Determine whether the following sets belong to W and, if yes, find their densities:
 - an arithmetic sequence with difference $d \in \mathbb{N}$;
 - the set of Fibonacci numbers;
 - the set of primes;
 - the set of indices of the primes of the form $4k + 3$ in the sequence of primes $p_1 = 2, p_2 = 3, p_3 = 5, p_4 = 7, \dots$
- e) Is the triple (X, W, p) admissible?
- f) Is (X, W) a Riemann pair?

4. Let the triple (X, W, p) be as in the previous question, and let $f : X \rightarrow \mathbb{R}$ be a bounded function. Determine interconnections between the following assertions (find out whether from any of them follows another):

- a) $\lim_{n \rightarrow +\infty} f(n)$ exists;
- b) $\lim_{n \rightarrow +\infty} \frac{f(1) + \dots + f(n)}{n}$ exists;
- c) the function f is Riemann integrable on (X, W, p) ;
- d) for any finite interval $]a, b[\subseteq \mathbb{R}$, the preimage $f^{-1}(]a, b[)$ belongs to W .

5. Check the properties a)–d) of the previous question for the functions $f(x) = \sin(cx)$, $f(x) = \arcsin(cx)$, $f(x) = \sin(\sin(cx))$, $f(x) = \sin(x^d)$ and $f(x) = \sin(d^x)$, where $c \in \mathbb{R}$ and $d \in \mathbb{N}$ are constants. Find the limits and the integrals if they exist.

6. Suggest and study additional directions of research.

7. Chromatic Polynomials of Numbered Graphs

Let $G = (V, E, \lambda)$ be a simple undirected labeled graph, where V is its vertex set, E is its edge set and $\lambda : E \rightarrow \mathbb{N}$ is a labeling of its edges with positive integers. A *proper vertex k -numbering* of G is a labeling $\nu : V \rightarrow \{1, \dots, k\}$ of its vertices with integers from 1 to k

satisfying the following condition: if any two vertices u and v are connected by an edge e , then

$$|\nu(u) - \nu(v)| \geq \lambda(e).$$

Note that it is not necessary that all the integers from 1 to k are used (see also Figure 5).

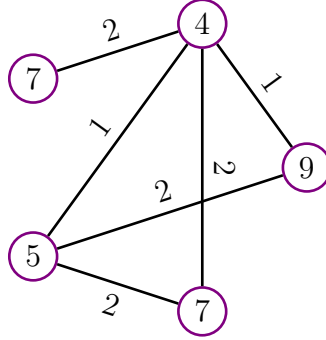


FIGURE 5. A proper vertex 9-numbering of a labeled graph.

Denote by $N_G(k)$ the number of proper k -numberings of a given labeled graph $G = (V, E, \lambda)$. In the questions 1-6 below, we additionally suppose that $\lambda(e) \in \{1, 2\}$ for all $e \in E$.

1. Prove that there exists a unique polynomial $p_G(t) \in \mathbb{Q}[t]$ with rational coefficients such that for some constant $k_0 \in \mathbb{N}$ (depending only on G and λ) and all $k \geq k_0$, one has $p_G(k) = N_G(k)$. We call $p_G(t)$ the *chromatic polynomial* of G .

2. Prove that $p_G(t) \in \mathbb{Z}[t]$, that is, the coefficients of $p_G(t)$ are in fact integers.

3. Show the following statements about $p_G(t)$:

- a) its degree equals the number $n = |V|$ of vertices of G ;
- b) its leading coefficient is 1;
- c) the coefficient at t^{n-1} equals $|E| - 2S(G)$, where $|E|$ is the number of edges of G and $S(G) = \sum_{e \in E} \lambda(e)$ is the sum of the edge labels of G .

4. Compute (as many as you can) the remaining coefficients of $p_G(t)$.

5. Denote by $G - e = (V, E \setminus \{e\}, \bar{\lambda})$ the graph obtained from G by *deleting* its edge e but preserving all the vertices and all other edges' labels: $\bar{\lambda}(h) = \lambda(h)$ for all $h \in E \setminus \{e\}$.

- a) Prove that if e is an edge with label 1, then the difference $p_{G-e}(t) - p_G(t)$ is actually the chromatic polynomial of another labeled graph H :

$$p_H(t) = p_{G-e}(t) - p_G(t).$$

Relate the graph H to the graph G .

- b) Try to find a similar relation for the edges with label 2.

6. Is it true that the coefficients of $p_G(t)$ alternate in sign, that is, $p_G(t) = \sum_{i=0}^n (-1)^i a_i t^{n-i}$ for some nonnegative integers $a_0, \dots, a_n$?

7. When is $p_G(t)$ irreducible over $\mathbb{Z}$?

8. Find other properties of chromatic polynomials.

9. Study the previous questions in a more general setting, for example, when the edge labels can be more than 2.

10. Suggest and study additional directions of research.

8. Lines over Finite Rings

Let $\mathbb{Z}_n = \{0, 1, \dots, n-1\}$ be the ring of residues modulo a positive integer n . For any two elements $a, b \in \mathbb{Z}_n$, we consider the set

$$L(a, b, n) = \{(x, y) \in \mathbb{Z}_n^2 \mid a \cdot x + b \cdot y = 0\}$$

and call it the *line over $\mathbb{Z}_n$ with coefficients a and b* .

1. Given n , what can the cardinality of a line over $\mathbb{Z}_n$ be?
2. Given n , find all possible values of the cardinality $|L(a_1, b_1, n) \cap L(a_2, b_2, n)|$ of the intersection of two lines over $\mathbb{Z}_n$.
3. Given n and $(a, b) \in \mathbb{Z}_n^2$, find the cardinality $|\{(a_1, b_1) \in \mathbb{Z}_n^2 \mid L(a_1, b_1, n) \subseteq L(a, b, n)\}|$, that is the number of the lines contained in the line $L(a, b, n)$.

Let $\mathbb{Z}_n[t]$ be the ring of polynomials with coefficients in $\mathbb{Z}_n$, and let $f(t) \in \mathbb{Z}_n[t]$ be a polynomial. Recall that $\mathbb{Z}_n[t]/(f(t))$ denotes the quotient ring of polynomials in which addition and multiplication are defined modulo $f(t)$. For example, $\mathbb{Z}_2[t]/(t^2 + t + 1)$ consists of four elements 0, 1, t and $t + 1$.

Put $R = \mathbb{Z}_n[t]/(f(t))$ and introduce, for any two elements $a(t)$ and $b(t)$ from R , the set

$$L_R(a, b, n) = \{(p, q) \in R^2 \mid a(t) \cdot p(t) + b(t) \cdot q(t) \equiv 0\}$$

that we will call the *line over R with coefficients a and b* .

4. Solve the problems similar to the questions 1, 2, and 3 above, when $f(t)$ is a polynomial of second degree and
 - a) n is prime;
 - b) n is any positive integer.
5. Suggest and study further generalizations and additional directions of research.

9. Decompositions of Integer Matrices

For any $n \times n$ real matrix A , there are well-known classical decompositions:

- *LU decomposition*: $A = LU$, where L is a lower triangular matrix, U is an upper triangular matrix.
- *Symmetric decomposition*: $A = S_1 S_2$, where S_1 and S_2 are two symmetric matrices.
- *Jordan-Chevalley decomposition*: $A = N + S$, where S is a semi-simple matrix, N is a nilpotent matrix and $SN = NS$.

Denote by $\mathbb{Z}^{n \times n}$ the set of all matrices with n rows and columns and integer entries (square integer matrices of dimension n).

1. Let $A \in \mathbb{Z}^{n \times n}$ be an integer matrix.
 - a) Find necessary and/or sufficient conditions on A so that there exists a decomposition of the above type (LU , symmetric or Jordan-Chevalley) into **integer** matrices.
 - b) Is such a decomposition unique? If not, describe all possibilities.
2. A matrix $B \in \mathbb{Z}^{n \times n}$ is called *prime* if $|\det(B)|$ is a prime number.
 - a) Is it true that, for every $A \in \mathbb{Z}^{n \times n}$ with $\det(A) \neq 0$, there exist (not necessarily different) prime matrices $B_1, \dots, B_k \in \mathbb{Z}^{n \times n}$ such that $A = B_1 \cdots B_k$?
 - b) Is such a decomposition unique? If not, describe all possibilities. Start with $n = 1, 2$.

A matrix $E \in \mathbb{Z}^{n \times n}$ is called *unit* if $|\det(E)| = 1$. We say that two integer matrices A and B are *weakly similar* if there exist units E and F such that $EAF = B$. We say they are *similar* if there is a unit E such that $EAE^{-1} = B$. We say that they are *strongly similar* if there exists a unit E which is orthogonal (that is, $E^T E = I_n$) and such that $EAE^{-1} = B$.

3. Is it true that every integer matrix is weakly similar to a diagonal matrix?
4. Describe all integer matrices that are similar (weakly similar, strongly similar) to a diagonal matrix.
5. Describe all integer matrices that are similar (weakly similar, strongly similar) to an upper triangular matrix.
6. Study existence and uniqueness of other classical decompositions (QZ -decomposition, singular value decomposition, etc.) for integer matrices.
7. Study similar questions for matrices with entries from the set of Gaussian integers $\mathbb{Z}[i] = \{a + bi \mid a, b \in \mathbb{Z}\}$.
8. Suggest and study additional directions of research.

10. Boundary Points of Special Spaces

A finite subset of natural numbers $F \subset \mathbb{N}$ will be called *good* if its cardinality isn't larger than its minimal element: $|F| \leq \min F$. A good set F is called *perfect* if $|F| = \min F$. For example, the set $\{5, 8, 13\}$ is good while the set $\{5, 8, 13, 14, 17\}$ is perfect.

A sequence $(x_k) = x_1, x_2, \dots, x_k, \dots$ of real numbers will be called *special* if the following two conditions are satisfied:

- (i) $\sum_{i \in F} |x_i| \leq 1$ for all good sets F ;
- (ii) $\sum_{i \in F} |x_i| = 1$ for at least one good set F (such a set is called *maximal* for the sequence).

Denote by $\mathcal{S}$ the space of all special sequences. An element $(x_k) \in \mathcal{S}$ will be called a *boundary point*, if from the equality

$$(x_k) = t \cdot (y_k) + (1 - t) \cdot (z_k) \text{ for some elements } (y_k), (z_k) \in \mathcal{S} \text{ and a real number } t \in]0, 1[$$

necessary follows that $(x_k) = (y_k) = (z_k)$.

1. Find all boundary points (x_k) of $\mathcal{S}$ with $x_k = 0$ for all $k > 6$.
2. Prove that for any boundary point (x_k) of $\mathcal{S}$ one has $x_1 = \pm 1$.
3. Prove that the following sequence is a boundary point of $\mathcal{S}$:

$$1, \frac{2}{5}, \frac{3}{5}, \frac{1}{5}, \frac{1}{5}, \frac{1}{5}, \frac{1}{5}, \frac{1}{5}, \frac{1}{5}, \frac{1}{5}, 0, 0, \dots$$

4. Let F be a maximal set for an element $(x_k) \in \mathcal{S}$. Suppose that $(x_k) = t \cdot (y_k) + (1 - t) \cdot (z_k)$ for some $(y_k), (z_k) \in \mathcal{S}$ and $t \in]0, 1[$. Show that F is also maximal for (y_k) and (z_k) .
5. Prove that, for any boundary point $(x_k) \in \mathcal{S}$ with finitely many non-zero terms, there exists exactly one maximal non-perfect set F such that $x_i \neq 0$ for all $i \in F$.
6. Prove that, for any boundary point $(x_k) \in \mathcal{S}$ with n nonzero terms, there exist at least n different maximal sets F such that $x_i \neq 0$ for all $i \in F$.
7. Does the space $\mathcal{S}$ have infinitely many boundary points? Try to find as many boundary points as possible.

8. Try to describe all boundary points of $\mathcal{S}$.
9. Let $c > 0$ be a rational number. A finite subset of natural numbers $F \subset \mathbb{N}$ will be called *c-good* if $|F| \leq c \cdot \min F$. Extend and study the problem for *c-good* sets instead of good sets.
10. A finite subset of natural numbers $F \subset \mathbb{N}$ will be called *square-good* if $|F| \leq (\min F)^2$. Investigate an analogous problem by considering square-good sets instead of good sets.
11. Suggest and study additional directions of research.

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